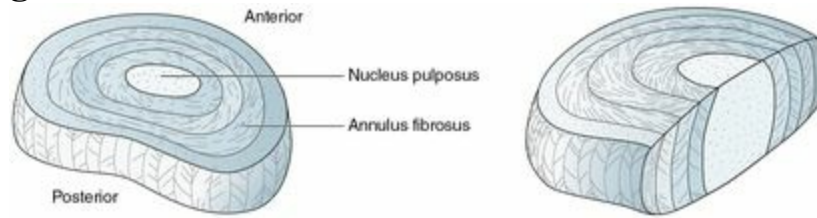


## Intervertebral Disc

The intervertebral disc has three major components: the nucleus pulposus, annulus fibrosus, and end plates. The nucleus has a gel-like character with collagen fibrils suspended in a base of water and various mucopolysaccharides, giving it both viscosity and some elastic response when perturbed in vitro. At the risk of sounding crude, the best way to describe a healthy nucleus is that it looks and feels like heavy phlegm. During in vitro testing, we have had it squirt out under pressure and literally stick to the wall. Although there is no distinct border between the nucleus and the annulus, the lamellae of the annulus become more distinct, moving radially outward. The collagen fibers of each lamina are obliquely oriented (the obliquity runs in the opposite direction in each concentric lamella). The ends of the collagen fibers anchor into the vertebral body with Sharpey's fibers in the outermost lamellae, whereas the inner fibers attach to the end plate (the end plate was discussed earlier). The discs in cross section resemble a rounded triangle in the thoracic region and an ellipse in the lumbar region, suggesting anisotropic facilitation of twisting and bending (see [figure 3.16](#)).

**Figure 3.16** Cross section of the intervertebral disc.



## Load-Bearing Abilities

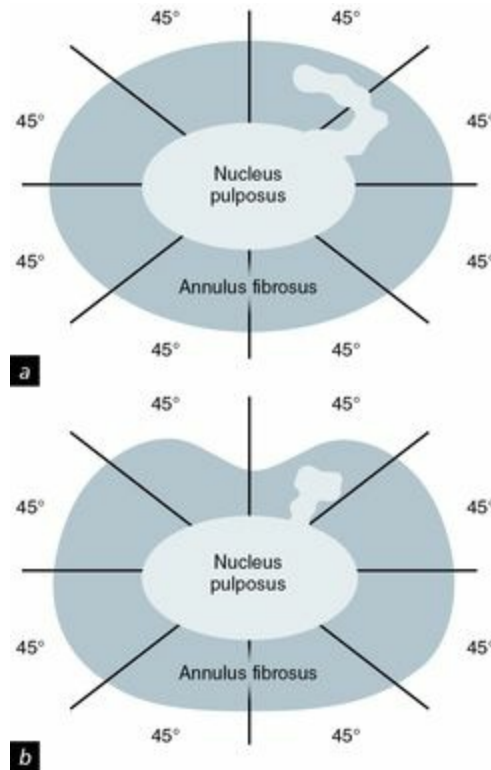
The disc behaves as a hydrostatic structure that allows  $6^\circ$  of motion between vertebrae. Its ability to bear load, however, depends on its shape and geometry. As a result of the orientation of the collagen fibers within the concentric rings of the annulus (one-half of the fibers are oblique to the other half), the annulus is able to resist loads when the disc is twisted. However, only half of the fibers can support this mode of loading; the other half become disabled, resulting in a substantial loss of strength or ability to bear load. The annulus and the nucleus work together to support compressive load when the disc is subjected to bending and compression. Under spine compression, the nucleus pressurizes, applying hydraulic forces to the end plates vertically and to the inner annulus laterally. This causes the annulus collagen fibers to bulge outward and become tensed. Years ago, Markolf and Morris (1974) elegantly demonstrated that a disc with the nucleus removed lost height but preserved much of its properties of axial stiffness, creep, and relaxation rates. The fact that the nucleus appears necessary to preserve disc height has implications for facet loading, shear stiffness, and ligament mechanics.

It is noteworthy that disc damage is most often accompanied by subdiscal bone damage (Gunning, Callaghan, and McGill, 2001). In fact, Keller and colleagues (1993) noted the interdependence of bone status and disc health. Evidence also suggests that excessive compression can lead to altered cell metabolism within the nucleus and increased rates of cell death (apoptosis) (Lotz and Chin, 2000). Thus, the evidence suggests that compressive loading involving lower compressive loads stimulates healthy bone (noted as a correlate of disc health), but that excessive loading leads to tissue breakdown.

The strength of the disc relative to the vertebrae was studied by Skrzpiec and colleagues (2007), who found age and gender effects. They found that the vertebrae lost 69% and 75% of compressive strength in males and females, respectively, from age 48 to 91 years. The annulus strength did not decrease except for the outermost layers in males' spines. The implication is that, given the minimal potential for adaptive change, the annulus is relatively weak in the strengthening spines of young men, but is relatively strong in the weakening spines of elderly females.

Other features of the disc influence load-bearing ability. First, larger discs develop more stress when bending and herniate more easily (Adams and Dolan, 2005). Yates, Giangregorio, and McGill (2010) found that disc shape determined the pattern of collagen delamination and herniation type (see [figure 3.17](#) a and b). Transverse sections reveal whether the disc is predominantly oval or limaçon shaped (shaped like a lima bean with the concave section around the posterior part adjacent to the spinal cord). The magnitude of stress concentrations in the annulus is a function of the radial distance from the neutral axis. This means that thicker discs experience more stress during bending and twisting. Because oval discs do not focus stress concentrations when bending or twisting, the patterns of the nucleus tracking through the annulus are diffuse. In contrast, the limaçon-shaped discs focus stress at the apex of the curve on either side of the concave section. This produces focal bulges posterolaterally (Yates, Giangregorio, and McGill, 2010). This type of bulge appears to respond better to direction-based therapy. There is a reason big athletes (with large-diameter limaçon-shaped discs) who generate very high stress can bear the stress without injury, but do poorly with activities such as yoga. Also, these athletes cannot hit a golf ball far because they bend and twist poorly. The archetypical golfer's spine is slender with ovoid discs that bend and twist with less stress, but it cannot survive high compressive loads. The higher-level implication is that the skilled clinician matches training and rehabilitation approaches with the anatomical features that govern function.

**Figure 3.17** Pathway of partial herniation in (a) an oval-shaped disc with volume exceeding an arc of 45°; (b) a limaçon-shaped disc directionally concentrated partial herniation with volume contained within an arc of 45°.



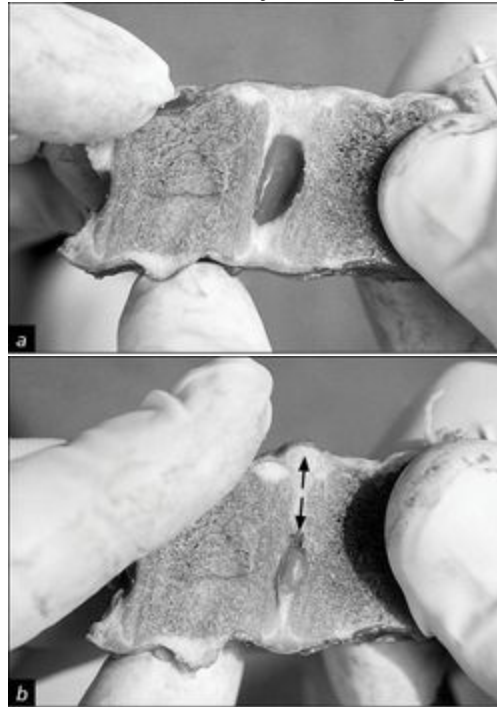
Adapted from Yates, Giangregorio, and McGill 2010.

## Progressive Disc Injury

Chapters 1 and 2 addressed the risk of disc injury as a function of occupation and other variables that are proxy measures of load on the disc. Here, a more detailed consideration of progressive disc injury is in order, focusing on primary variables that damage tissue rather than surrogate descriptors. A normal disc under compression deforms mainly the end plates vertically and displays outward bulging of the annulus (Brinckmann, Biggemann, and Hilweg, 1988). However, if little hydrostatic pressure is present, as in the case in which the nucleus has been lost through end-plate fracture or herniation, the outer annulus bulges outward and the inner annulus bulges inward during disc compression (see [figure 3.18](#), a and b). This double-convex bulging causes the laminae of the annulus to separate, or delaminate, and has been hypothesized as a pathway for nuclear material to

leak through the lamellae layers and finally extrude, creating a frank herniated disc (Adams and Dolan, 1995).

**Figure 3.18** (a) A healthy joint with a contained nucleus has minimal annulus deformation under compressive load. (b) If the nucleus loses pressure (as a result of an end-plate fracture, for example), the annulus compresses, causing radial bulging both outward and inward (arrows) and producing delaminating stresses as the annulus layers are pulled apart.



Photos from Stuart McGill

Professor Michael Adams put forth an interesting proposal (personal communication, June, 2015). He suggested that a healthy disc builds internal pressures under compressive loads that are so high that no nerve or vascular vessel could survive. Following initial end-plate damage, the disc can no longer build substantial pressures so that nerves and blood vessels are able to invade the disc. These are more possibilities to explain the increased vascularization of so-called degenerated discs and their ability to generate pain.

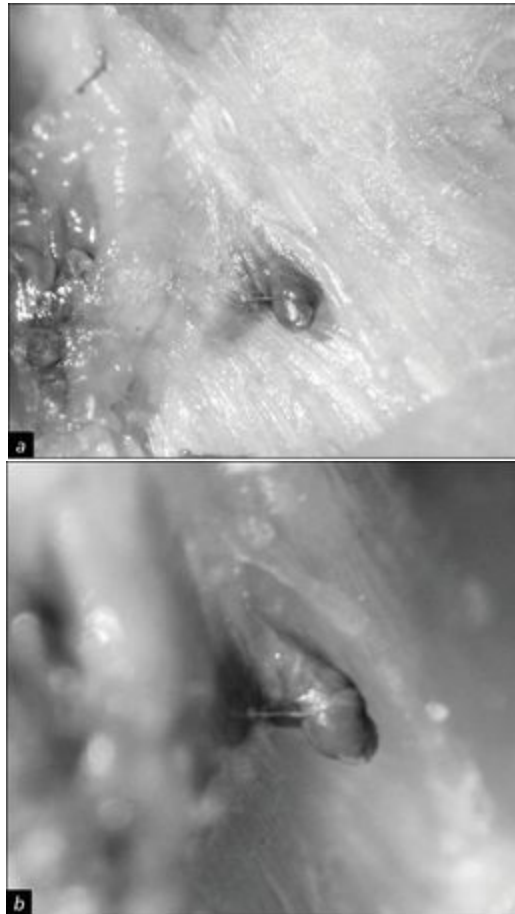
The following two sections review factors that influence disc damage and those that influence recovery.

## **Influences on Disc Damage**

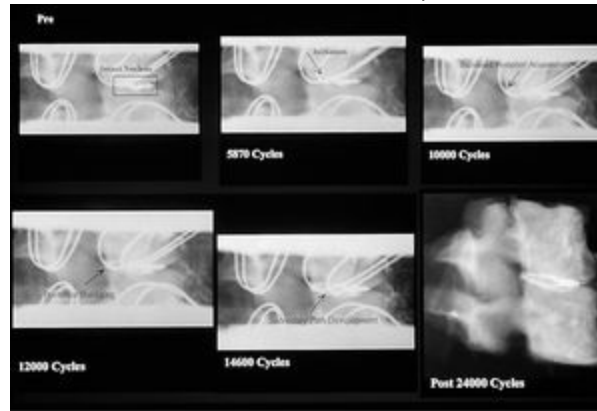
Several features of disc herniation tie together injury mechanisms and

therapy. Most herniations have no tearing of the collagen fibers. Rather, the fibers delaminate. They do this in two ways. First, with repeated bending, adjacent fibers within the same layer eventually separate allowing the nucleus gel to flow into this cleft (Tampier et al., 2007). Bending also creates a hydrostatic pressure concentration on these fibers on the outside of the bend (i.e., on the posterior fibers when bending forward). Repeated bending causes this delamination process to continue with the next concentric annulus layer (see [figure 3.19](#)). In this way, this type of damage works from the inside out (Veres, Robertson, and Broom, 2009). Time series radiographs (see [figure 3.20](#)) demonstrated the progression of the nuclear material tracking through the annulus with successive motion cycles. The herniated disc appears to result from cumulative trauma: even though we have crushed several thousand vertebral motion segments, we have only once or twice observed a herniation without concomitant flexion cycles. Note that we are including both frank herniation and visible disc bulges (see [figure 3.21](#)) under this category of injury mechanism.

**Figure 3.19** (a) Peeling off some collagen layers reveals the nucleus tracking through the layers of the annulus from the inside out. (b) Removing another layer of the annulus shows how the nucleus tracks through the annulus.



**Figure 3.20** Serial radiographs showing the initial containment of the radiocontrast in the nucleus and, with repeated full-flexion motion (with about 1,400 N of compression), the progressive posterior tracking of the contrast until frank herniation after 24,000 bending cycles.



Reprinted from Clinical Biomechanics, 16(1), J.P. Callaghan and S.M. McGill, "Intervertebral disc herniation: Studies on a porcine model exposed to highly repetitive flexion/extension motion with force," 28-37, 2001, with permission from Elsevier Science.

**Figure 3.21** Repeated flexion can result in a frank herniation or a bulge, shown here in a specimen with the lamina removed to expose the posterior disc, which can impinge on nerves, causing symptoms. Note that the nucleus was injected with a dye so that the bulge is more visible (arrow).



Photo from Stuart McGill

As noted earlier, disc size and shape influence the rate of this progress (Yates, Giangregorio, and McGill, 2010). I also strongly suspect from our animal work that repeated bending softens the collagen matrix so that repeated stretching and bending allows this delamination process to occur even faster. Although several reports have documented broken end-plate pieces extruded with the nucleus material, our work has reported microfracturing of the end plate where the annulus collagen fibers attach (Balkovec et al., 2014). Thus, although the fibers do not appear to rupture along their length, they can break away from their connection to the end plate. This is further evidence that herniation may often begin with an overload of compressive force and finish with a bending cycle. The excessive compression pressurizes the nucleus to the point that stress on the end plate causes cracks within the plate (Brown, Gregory, and McGill 2008) and breaking of the trabecular bone underneath (Balkovec et al., 2014).

This process is initiated and progressed with repeated bending. Additional compressive load accelerates the process. Interestingly we have not been able to initiate the delamination process with prolonged flexion, as would occur with sitting. Rather, once the delamination has begun with bending (probably full flexion; Adams and Hutton, 1982), prolonged flexion exacerbates the flow of nucleus into the delaminated clefts. This means that sitting probably does not initiate disc injury, but once bending initiates the damage, sitting can

become painful. This helps to make sense of the epidemiological data that link herniation with sedentary occupations and the sitting posture (Videman, Nurminen, and Troup, 1990).

Several years ago we sought the most potent mechanism leading to disc herniation. We found that repeated flexion motion under simultaneous compressive loading was the easiest way to ensure herniation. In fact, it turned out that the number of cycles of flexion motion was more important than the actual magnitude of compressive load. Although no herniations were produced with 260 N of compressive load and up to 85,000 flexion cycles, herniations were produced with 867 N of load and 22,000 to 28,000 cycles, and with 1,472 N and only 5,000 to 9,500 cycles (Callaghan and McGill, 2001). Although people vary greatly, these numbers illustrate the relationship: herniations are a function of repeated full-flexion motion cycles with only a modest level of accompanying compressive load. In fact, we mimicked the lumbar motion and loading of a typical spine rehabilitation machine in which the seated patient belts down the pelvis to isolate the lumbar spine and then extends the torso repetitively against a resistance over the midback. (Amazingly, some people are trained on this type of machine, even those with known disc herniations!) (See [figure 3.22](#).) Logically, one could follow a McKenzie-type exercise philosophy, believing that the extension motion can reduce a posteriorly displaced nucleus (we observed this in some specimens; Scannell and McGill, 2009). Yet, more recent work (Balkovec and McGill, 2012) in which we measured the accelerated rate of disc flattening with repeated flexion–extension cycles motivated us to seek better alternatives.

**Figure 3.22** Rehabilitation devices such as this attempt to isolate lumbar motion by extending against a resistance pad but create simultaneous lumbar compression. We found that replicating the full range of motion from full flexion to neutral and using the compressive loads of these devices was a powerful combination that produced disc herniation.



Other aspects of our work have revealed the dependency of the location of the herniating bulge on the axis of motion (Aultman, Scannell, and McGill, 2005). For example, in 20 motion segments, we flexed them repeatedly about an axis that was  $30^\circ$  rotated from the pure flexion axis (mostly flexion with some lateral bend). One specimen simply failed abruptly and was removed. In the remaining 19, the herniation track was away from the axis of rotation (see [figure 3.23](#)).

**Figure 3.23** Tilting the flexion axis 30° away from pure flexion caused the nucleus to track in a direction away from the axis (Aultman, Scannell, and McGill, 2005). This motion dependence has powerful potential for the design of exercise for those with known posterolateral disc bulges.

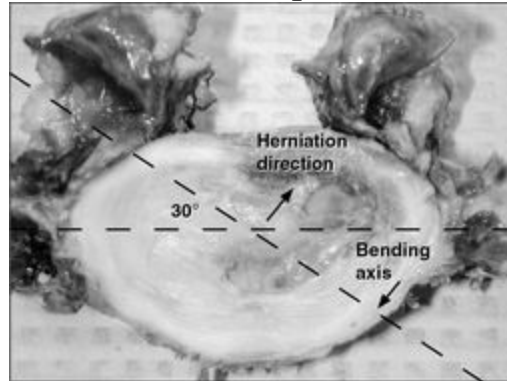


Photo from Stuart McGill

Herniations tend to occur in younger spines (Adams and Hutton, 1985), meaning those with higher water content (Adams and Muir, 1976) and more hydraulic behavior. Older spines do not appear to exhibit classic extrusion of nuclear material, but rather, are characterized by radial cracks that appear to progress with repeated loading (a nice review is provided by Goel, Monroe, et al., 1995).

Twisting initiates a different type of collagen delamination than with pure bending. Here twisting causes stress between the adjacent rings of collagen. More twisting cycles causes stress-strain reversals and clefts between the layers ([figure 3.24](#)). Some radiologists report these as disc tears, but we recognize them as subsequent to twisting damage (see [figure 3.25](#) a and b). Thus, both forms of delamination, between fibers and between layers, can be classified as cumulative trauma damage. Although we do not yet know the relationship between the number of cycles and loads, we do know that added torsion reduces the compressive strength of the joint (Aultman et al., 2004).

**Figure 3.24** Twisting causes concentric layers of the annulus to delaminate. Here a layer has been dissected away to reveal the tracking of the nucleus between annulus layers.

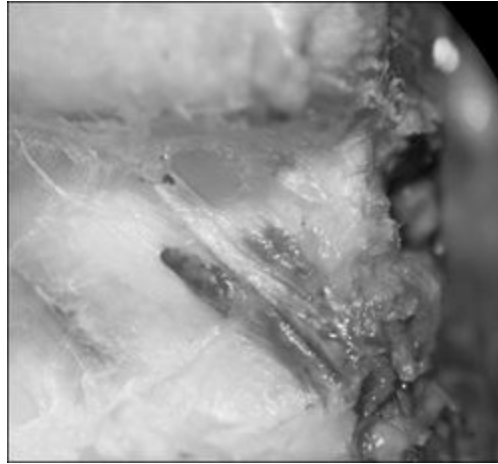


Photo from Stuart McGill

**Figure 3.25** Delamination patterns from repeated twisting. The nucleus was dyed to show separation of concentric rings of collagen. (a) Complete circumferential delamination; (b) partial circumferential delamination.

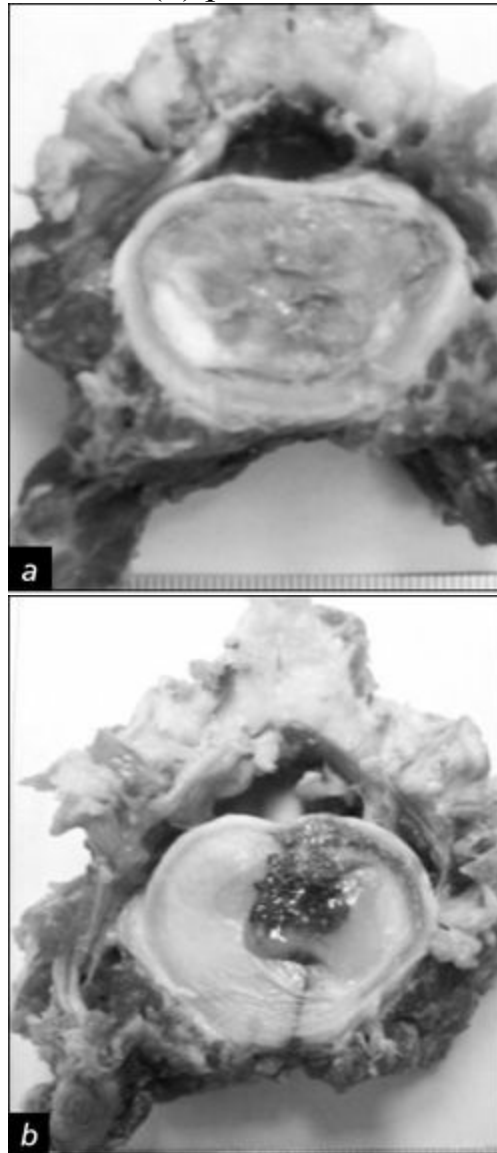


Photo from Stuart McGill

## **Mechanisms of Annulus Failure (Herniation)**

### **Clinical Relevance**

Damage to the annulus of the disc (herniation) appears to be associated with fully flexing the spine for a repeated or prolonged period of time. In

fact, herniation of the disc seems almost impossible without full flexion, and without some compressive load. This has implications for exercise prescription particularly for flexion stretching and sit-ups or for activities such as prolonged sitting, all of which are characterized by a flexed spine. Some resistance exercise machines that take the spine to full flexion repeatedly must be reconsidered for those interested in sparing the posterior annulus portions of their discs. Furthermore, the mechanism by which the process can be interrupted appears to be postural dependent, providing some more insight into the mechanism of the McKenzie approach. These summary thoughts provide a basis for the prescription of the cat–camel exercise, for example, as a preferred position for moving the spine. Being on the hands and knees removes the compressive load.

Also worth noting here is the intriguing hypothesis of Bogduk and Twomey, who suggested the possibility of an annular sprain similar to a sprain of the ankle ligaments (Bogduk and Twomey, 1991). They hypothesized that the outer layers of the annulus experience excessive strain under torsion. Given that these authors have presented evidence for the presence of nerve fibers particularly in this region, the annulus appears to be a good candidate for a source of pain.

## **Influences on Disc Recovery**

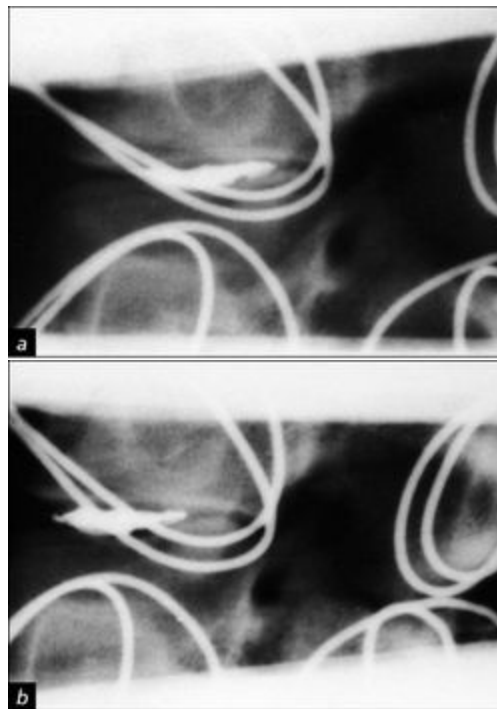
That disc bulges shrink over time is quite well documented. Benson and colleagues (2010) showed that the overwhelming majority of massive disc bulges shrink over a period of 2 years, as do the symptoms (83% had a complete and sustained resolution of symptoms, and volumetric analysis showed that bulges had reduced 64% in size on average). Several possibilities for a shrinking mechanism exist. The extruded material may be resorbed by the disc or may dehydrate. The inflammatory response may cause digestion of the extruded nucleus. Miyazaki and colleagues (2009) documented the ingrowth of vascular tissues into the extruded material and believed that this mechanism brings macrophages that digest the foreign body. However, subsequent work showed that vascularization was accompanied by granulation tissues on the dura and nerve roots. They linked this clinically to radiating symptoms that disrupted nerve conduction and reduced the pain-

free range during the straight-leg raise (Kobayashi et al., 2010). This understanding is the underpinning of some of the nerve flossing techniques introduced later.

We have also been able to add more insight into the McKenzie therapy approach, which is based on extended postures. The theory is that an extended posture drives the nucleus forward within the disc, and quite possibly vacuums in the partially extruded nucleus. As the layers of the annulus delaminate and fill with nucleus material, we now have proof that extended postures can drive the nucleus material that is in the delaminated pockets of the posterior nucleus back toward the central part of the disc (Scannell and McGill, 2005) (see [figure 3.26](#)). Physical therapists have claimed that movements such as the floppy push-up milk the nucleus back to the center of the disc. Scannell showed this to be true, but only in discs with 70% of their original disc height remaining. But I am concerned over the trauma to the facets in this dynamic therapy, which results in an extension-intolerant facet patient a couple of years later. Subsequently, we tested spines in a static extended posture (prone with a fist under the chin). It appears that while in this posture, the nucleus material in the posterior annulus slowly returns to the center of the disc, but avoids the trauma to the facets. Once again, it appears to be possible only in discs with 70% of their height remaining. We need to further refine this work.

**Figure 3.26** Proof that disc bulges can be reduced with extended postures that appear to vacuum the nucleus that has invaded the posterior annulus.

This worked only in discs with 70% of their original height, or more, remaining. (a) The nucleus is working posteriorly with repeated flexion motion; (b) extension reverses the direction of the leading edge of the nucleus.



Photos from Stuart McGill

The technique of repeated dynamic extension movements to reduce displaced nucleus in the annulus causes additional concern. Balkovec and McGill (2012) compared repeated flexion of the spine with repeated flexion and extension. The hypothesis tested was that the extension component of each cycle could prevent the posterior migration of the nucleus. We observed that the added extension simply mashed down the disc faster. In other words, adding repeated extension greatly increased the rate of disc deterioration. We retain our position in favor of static extension postures for acute posterior disc bulges with 70% of the disc height remaining, and discourage dynamic extension movements such as the floppy push-up and the prone press-up.

Finally, some therapists advocate flexing discs postsurgery and postinjury because they believe that this reduces the risk of scar tissue formation and adhesions. One could argue that scarring could be desirable to form a plug of the nucleus leaking in the outer layers of the annulus. Obviously, scars

tethering neural tissues are problematic (more on this in the nerve flossing discussion). Adams, Stefanakis, and Dolan (2010) presented an excellent discussion of the many factors that influence the potential for healing. For these reasons we often advocate the cat–camel exercise performed while on the hands and knees. We have shown that minimal load is placed on the disc during the cat–camel given the horizontal orientation, and that bending without compressive load does not appear to exacerbate a disc bulge.

## **Muscles**

Traditional anatomical descriptions of the spine musculature have taken a posterior vantage point. This has hindered insight into the roles of these muscles because many of the functionally relevant aspects are better viewed in the sagittal plane. (For a nice synopsis of the sagittal plane lines of action, see Bogduk, 1980, and Macintosh and Bogduk, 1987.) Furthermore, many have developed their understanding of muscle function by simply interpreting the lines of action and region of attachment, assuming that the muscles act as straight-line cables. This may be misleading.

Muscles of the torso are fundamentally different from those of the limbs from a motor control perspective. Limb muscles create motion; torso muscles more often stop or control motion. There really are no agonists or antagonists in the torso because all of the muscles are required to stabilize and control motion generated elsewhere. Studies of athletic performance show that function is optimized when power is generated at the hips and transmitted through a stiffened core, or torso, with no energy leaks. This has a large impact on the approach to training torso muscle for both rehabilitation and performance enhancement. An examination of the architecture and mechanics of the abdominal wall, for example, shows that the wall uses the three layers as a structural composite to enhance stiffness and produce hoop stresses. The rectus abdominis assists the production and transmission of hoop stresses around the torso via the lateral tendons. Thus, training the muscle structures through the range of motion as in a sit-up, for example, would not address one of the primary functions.

## **Location of Disc Bulge and Designing Exercise**

### **Clinical Relevance**

A posterolateral disc bulge is usually caused by repeated flexion of the disc about an axis cutting across the disc perpendicular to the herniating track. This is powerful knowledge for exercise intervention, because further motion about this axis would exacerbate the herniation. This knowledge gives clues for better prevention. Look for a dominant motion pattern in a patient's daily routine that is consistent with the bulge location, and eliminate it. If the

causative motion pattern is an element of an athletic event in which the patient competes, major decisions will need to be made. More motion will only ensure the inevitable. Can the technique be changed to eliminate the causative motion?

A different approach is required for a full understanding of the function and purpose of each muscle. First, knowledge of static muscle morphology is essential, although it may change over a range of motion. Second, knowledge of activation-time histories of the musculature must be obtained over a wide variety of movement and loading tasks. Muscles create force (and stiffness), but these forces play roles in moment production for movement and in stabilizing joints for safety and performance. A further understanding of the motor control system strategies chosen to support external loads and maintain stability requires the interpretation of anatomy, mechanics, and activation profiles. This section enhances the discussion of anatomically based issues of the spine musculature and blends the results of various electromyographic (EMG) studies to help interpret function and the functional aspects of motor control.

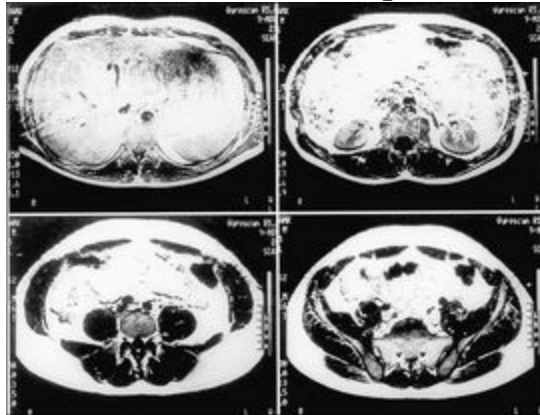
## **Muscle Size and Force Production**

The physiological cross-sectional area (PCSA) of muscle determines the force-producing potential, whereas the line of action and moment arm determine the effect of the force in moment production, stabilization, and so forth. It is erroneous to estimate muscle force based on muscle volume without accounting for fiber architecture or by taking transverse scans to measure anatomical cross-sectional areas (McGill, Patt, and Norman, 1988). Using areas directly obtained from computed tomography (CT) or magnetic resonance imaging (MRI) slices has led to erroneous force estimates and interpretations of spine function. In such cases, because a large number of muscle fibers are not seen in a single transverse scan of a pennated muscle, muscle forces are underestimated. Thus, areas obtained from MRI or CT scans must be corrected for fiber architecture and scan plane obliquity (McGill, Santaguida, and Stevens, 1993).

In [figure 3.27](#), transverse scans of one subject show the changing shape of the torso muscles over the thoracolumbar region, highlighting the need to

combine transverse scan data with data documenting fiber architecture obtained from dissection. In this example, the thoracic extensors (longissimus thoracis and iliocostalis lumborum) seen at T9 provide an extensor moment at L4, even though they are not seen in the L4 scan. Only their tendons overlie the L4 extensors. A sagittal plane schematic shows the errors in measuring muscle cross-sectional area from a single transverse slice—which has caused some to underestimate the potential of the muscle.

**Figure 3.27** Transverse scans of one subject (supine) at the levels of T9, L1, L4, and S1, showing the musculature in cross section. Note that many muscles seen at more superior levels pass tendons over the lower levels, transmitting force. This illustrates the error in using a single scan to estimate muscle force and moment potential.



Reprinted from *Clinical Biomechanics*, Vol 8(4), S.M. McGill, L. Santaguida, and J. Stevens., “Measure of the trunk musculature from T6 to L5 using MRI scans of 15 young males corrected for muscle fiber orientation,” 171, 1993, with permission from Elsevier Science.

Raw muscle PCSAs and moment arms (McGill, Santaguida, and Stevens, 1993) are provided in appendix A.1. Areas corrected for oblique lines of action are shown in [table 3.1](#) for selected muscles at several levels of the thoracolumbar spine. Guidelines for estimating true physiological areas are provided in McGill, Patt, and Norman (1988). Marras and colleagues (2001) is another recent source of raw muscle geometry obtained from MRI for both males and females.

**Table 3.1** Corrected Muscle Cross-Sectional Areas

| Muscle*                   |       | Cross-sectional area (mm) <sup>2</sup> | Moment arms (anteroposterior) (mm) | Moment arms (lateral)(mm) |
|---------------------------|-------|----------------------------------------|------------------------------------|---------------------------|
| Longissimus pars lumborum | L3-L4 | 644                                    | 51                                 | 17                        |
| Quadratus lumborum        | L1-L2 | 358                                    | 31                                 | 43                        |
|                           | L2-L3 | 507                                    | 32                                 | 55                        |
|                           | L3-L4 | 582                                    | 29                                 | 59                        |
|                           | L4-L5 | 328                                    | 16                                 | 39                        |
| External oblique          | L3-L4 | 1,121                                  | 17                                 | 110                       |
| Internal oblique          | L3-L4 | 1,154                                  | 20                                 | 89                        |

\*The laminae of longissimus pars lumborum at the L4-L5 level would have been listed here by virtue of their cosines, but were not, because they could not be distinguished on all scan slices.

A few examples of corrected cross-sectional areas, anteroposterior moment arms, and lateral moment arms perpendicular to the muscle fiber line of action using the cosines are listed in McGill, Patt, and Norman (1988). These are the values that should be used in biomechanical models rather than the uncorrected values obtained directly from scan slices.

Moment arms of the abdominal musculature are generally obtained from CT- or MRI-based studies. Generally, the subject lies supine or prone within the MRI or CT scanner, and the distance from the spine to the muscle centroid is measured. Lying on the back in this posture causes the abdominal contents to collapse posteriorly under gravity (McGill, Juker, and Axler, 1996). In real life and during standing, the abdominal muscles are pushed away from the spine by the visceral contents. Recently, research has shown that CT or MRI studies of the abdominal muscle moment arms obtained from subjects in the supine posture underestimated the true values by 30%.

Other variables influence muscle force and stiffness production. Stress describes the amount of force per unit of muscle cross-sectional area. Generally, we have found in our modeling work that this value resides within the range of 35 to 60 N/cm<sup>2</sup>. Highly trained people would be in the higher end of the range. What we don't know is the difference between individual muscles. Again, given our modeling work, we strongly suspect that the oblique muscles produce more force per cross-sectional area than more bulky, traditionally shaped muscles. Another variable recently investigated is the ultrastructural design in terms of sarcomere length. Ward and colleagues (2009a) noted that the multifidus, for example, has shorter sarcomeres than any other muscle they studied in the body. This means that the multifidus is stiffer and creates more force at lengths longer than rest length. Fiber type and muscle spindle concentration are yet other variables that modulate force production. Although the erector spinae muscles are generally considered to

have a higher proportion of less-fatigable Type I muscle fibers, some have suggested that people with back pain are more fatigable and have fewer Type I fibers. Although on average, people with back pain are more easily fatigued, the evidence suggests that their fiber type distribution is no different from that of people who are less easily fatigued (Crossman et al., 2004).

Muscle spindles sense length and modulate force output accordingly. For example, if a muscle is elongated, the spindles instigate a pulse of force via the spindle-driven stretch reflex. Professor Pickar and colleagues studied spindle behavior and noted that prior length history (such as stretching) changes force production. For example, short, static stretches reduced subsequent spindle responsiveness (Ge and Pickar, 2012). This explains some of the performance losses following sitting, for example, which is revisited later in part II on injury prevention.

In summary, understanding the force and mechanical potential of muscles requires an appreciation for the curving line of action, which is best obtained in the anatomy lab. But, unfortunately, these specimens are usually atrophied, eliminating them as a source for muscle size estimates. Muscle areas obtained from medical imaging techniques need to be corrected to account for fiber architecture and contractile components that do not appear in the particular scan level (e.g., only the tendon passes the level). Further, moment arms for muscle lines of action from subjects who are lying down need to be adjusted for the application to upright postures maintained in real life. Thus, several variables influence force and stiffness production, which in turn influences spine stability, performance, and pained conditions. Finally, there is also the tendency to classify muscles as flexors, extensors, and the like. Every muscle is a stabilizer, a proximal stiffener that enables distal motion, and a stiffener of the flexible spine to enable load bearing without buckling. For this reason the muscles are listed in sections as anterior, posterior, and so on.

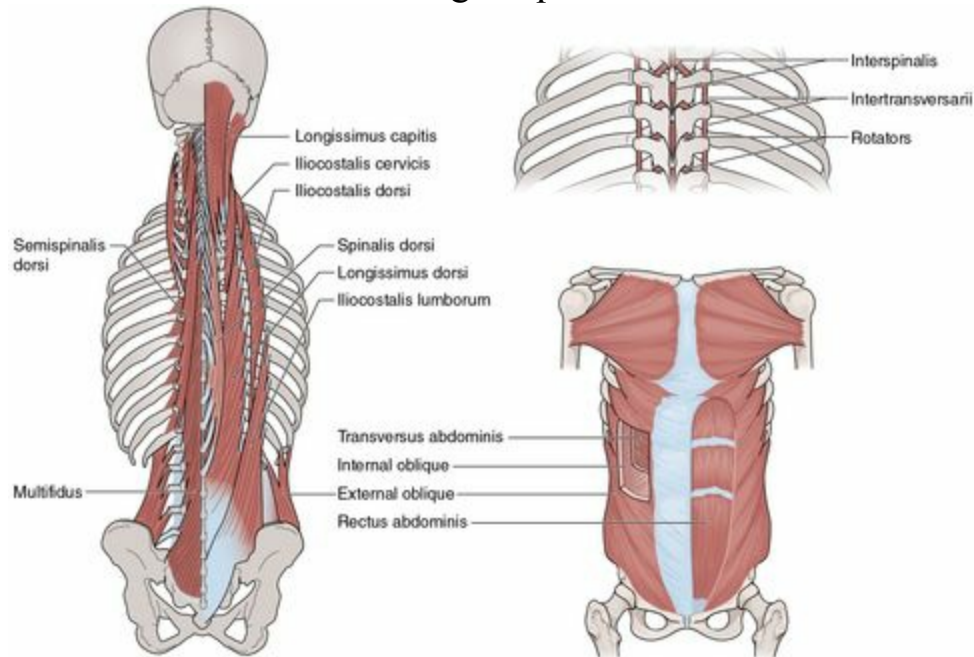
## **Posterior Muscle Groups**

This section describes muscle groups from a functional perspective and introduces some issues fundamental for understanding injury avoidance and the choice of appropriate rehabilitation approach.

### **Rotatores and Intertransversarii**

Many anatomical textbooks describe the function of the small rotator muscles of the spine, which attach to adjacent vertebrae, as creating axial twisting torque. This is consistent with their nomenclature (rotatores). Similarly, the intertransversarii are often assigned the role of lateral flexion. These proposals have several problems. First, these small muscles (see [figure 3.28](#)) have such small PCSAs that they can generate only a few newtons of force, and second, they work through such a small moment arm that their total contribution to rotational axial twisting and bending torque is minimal. For these reasons, I believe they serve another function.

**Figure 3.28** Short muscles of the spine: levator costarum longi, levator costarum brevi, intertransverse medialis, intertransverse lateralis, rotatores, and interspinalis. They have been described erroneously as creating axial twisting torque.



The rotatores and intertransversarii muscles are highly rich in spindles, approximately 4.5 to 7.3 times richer than the multifidus (Nitz and Peck, 1986). This evidence suggests that they may function as length transducers or vertebral position sensors at every thoracic and lumbar joint. In some EMG experiments we performed a number of years ago, we placed indwelling electrodes very close to the vertebrae. In one case we had a strong suspicion that the electrode was in a rotator muscle. The subject attempted to perform isometric twisting efforts with the spine untwisted (or constrained in a neutral posture) in both directions but produced no EMG activity from the rotator—only the usual activity in the abdominal obliques and so on. However, when the subject tried to twist in one direction (with minimal muscular effort), there was no response, whereas in the other direction there was major activity. This particular rotator seemed not to be activated to create axial twisting torque but rather in response to twisted position change. Thus, its activity was elicited as a function of the twisted position—which was not consistent with the assumed role of creating torque to twist the spine. This is stronger evidence that these muscles are not rotators at all but function as

position transducers in the spine proprioception system.

## **Manual Therapy and the Function of the Rotatores and Intertransversarii**

### **Clinical Relevance**

We now suspect that the rotatores and intertransversarii are actually length transducers and thereby position sensors, sensing the positioning of each spinal motion unit. These structures are likely affected during various types of manual therapy with the joint at its end range of motion (a posture used in chiropractic technique, for example).

### **Longissimus, Iliocostalis, and Multifidus Groups**

The major extensors of the thoracolumbar spine are the longissimus, iliocostalis, and multifidus groups. Although the longissimus and iliocostalis groups are often separated in anatomy books, it may be more enlightening in a functional context to recognize the thoracic portions of both of these muscles as one group and the lumbar portions as another group. The lumbar and thoracic portions are architecturally (Bogduk, 1980) and functionally different (McGill and Norman, 1987). Bogduk (1980) partitioned the lumbar and thoracic portions of these muscles into longissimus thoracis pars lumborum and pars thoracis, and iliocostalis lumborum pars lumborum and pars thoracis.

These two functional groups (pars lumborum, which attach to lumbar vertebrae, and pars thoracis, which attach to thoracic vertebrae) form quite a marvelous architecture for several reasons and are discussed in a functional context with this distinction (i.e., pars lumborum versus pars thoracis). Fiber typing studies note differences between the lumbar and thoracic sections: the thoracic sections contain approximately 75% slow-twitch fibers, whereas the lumbar sections are generally evenly mixed (Sirca and Kostevc, 1985). The pars thoracis components of these two muscles attach to the ribs and vertebral components and have relatively short contractile fibers with long tendons that run parallel to the spine to their origins on the posterior surface of the sacrum

and medial border of the iliac crests (see [figure 3.29](#)). Furthermore, their line of action over the lower thoracic and lumbar region is just underneath the fascia, such that forces in these muscles have the greatest possible moment arm and therefore produce the greatest amount of extensor moment with a minimum of compressive penalty to the spine (see [figure 3.30](#)). When seen on a transverse MRI or CT scan at a lumbar level, pars thoracis tendons have the greatest extensor moment arm, overlying the lumbar bulk—often over 10 cm (4 in.) (McGill, Patt, and Norman, 1988; McGill, Santaguida, and Stevens, 1993) (see [figure 3.31](#)).

**Figure 3.29** An isolated bundle of longissimus thoracis pars thoracis (inserting on the ribs at T6), with tendons lifted by probes, course over the full lumbar spine to their sacral origin. They have a very large extensor moment arm (just underneath the skin).

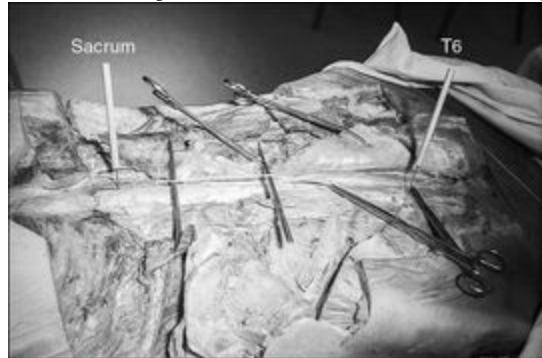


Photo from Stuart McGill

**Figure 3.30** This world-class powerlifter exemplifies the hypertrophied bulk of the iliocostalis and longissimus muscles seen in trained lifters. This muscle bulk is in the thoracic region, but the tendons span the entire lumbar spine.

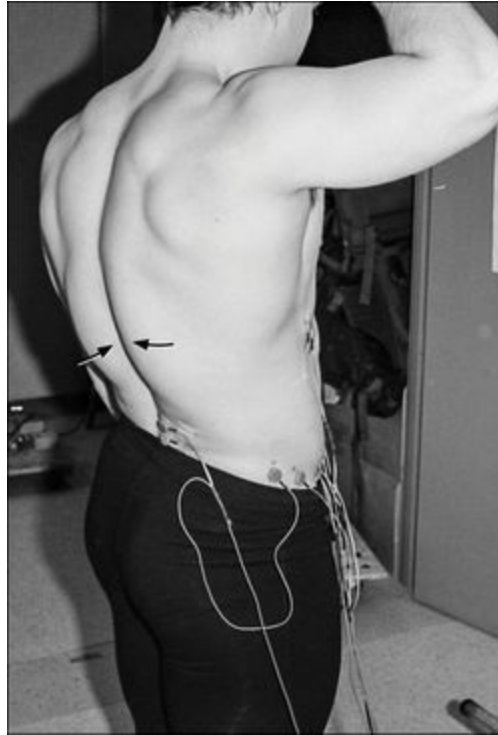
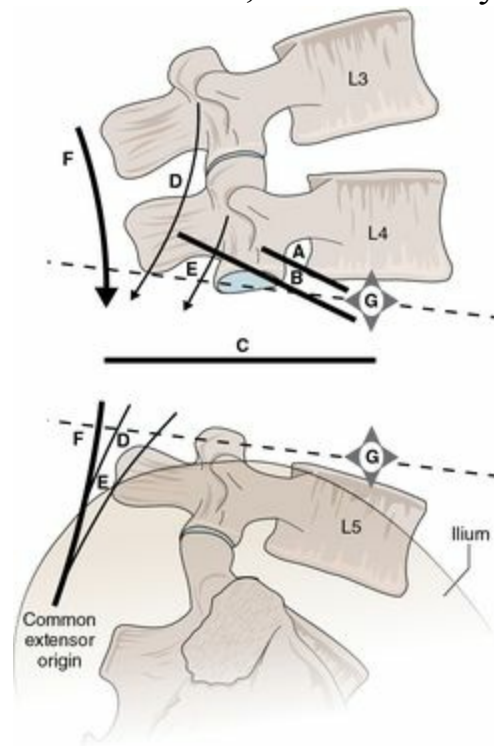


Photo from Stuart McGill

**Figure 3.31** (A, B) The moment arms of the pars lumborum portions of iliocostalis and longissimus. (C) The large moment arm of the pars thoracis iliocostalis and longissimus muscles gives them their ability to create major lumbar extensor moments. The lines of action of (D) the pars lumborum portion of longissimus, (E) the pars lumborum portion of iliocostalis, and (F) the pars thoracis portion of longissimus and iliocostalis join at their common point of origin at the sacral spine. (G) The mechanical fulcrum, or the axis about which muscles create moments, is indicated by the diamond shape.



The lumbar components of these muscles (iliocostalis lumborum pars lumborum and longissimus thoracis pars lumborum) are very different anatomically and functionally from their thoracic namesakes. They connect to the mamillary, accessory, and transverse processes of the lumbar vertebrae and originate, once again, over the posterior sacrum and medial aspect of the iliac crest. Each vertebra is connected bilaterally with separate laminae of these muscles (see [figure 3.32](#)). Their line of action is not parallel to the compressive axis of the spine but rather has a posterior and caudal direction that causes them to generate posterior shear forces together with extensor moment on the superior vertebrae (see [figure 3.33](#)). These posterior shear forces support any anterior reaction shear forces of the upper vertebrae that

are produced as the upper body is flexed forward in a typical lifting posture. It is important to clarify that this flexion of the torso is accomplished through hip rotation, not lumbar flexion. These muscles lose their oblique line of action and reorient to the compressive axis of the spine with lumbar flexion (McGill, Hughson, and Parks, 2000), so that a flexed spine is unable to resist damaging shear forces (see [figure 3.34](#), a-d). This possible injury mechanism, together with activation profiles during clinically relevant activities, is addressed in a later section.

**Figure 3.32** Iliocostalis lumborum pars lumborum and longissimus thoracis pars lumborum originate over the posterior surface of the sacrum, follow a very superficial pathway, and then dive obliquely to their vertebral attachments. This oblique orientation creates posterior shear (S) forces and extensor moments on each successive superior vertebra. The compressive axis (C) is indicated.

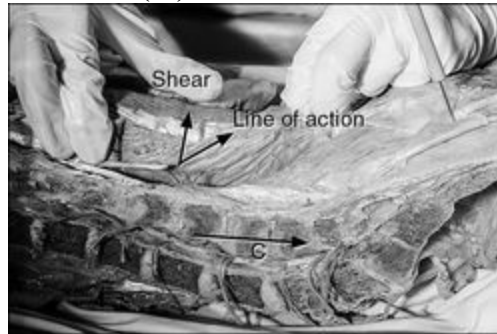


Photo from Stuart McGill

**Figure 3.33** The oblique angle of the lumbar portions of longissimus and iliocostalis is seen in vivo in this MRI picture. Their line of force (F) is shown relative to the compressive axis (C).



Photo from Stuart McGill

**Figure 3.34** The oblique angle of the lumbar portions of the iliocostalis lumborum and longissimus thoracis protects the spine against large anterior shear forces. However, this ability is a function of spine curvature. (a) A neutral spine and (b) the oblique angle of these muscles as viewed with an ultrasound imager. (c) The loss of this angle with spine flexion (d) so that anterior shear forces cannot be counteracted. This ensures shear stability and is another reason to consider adopting a neutral spine during flexed weight-holding tasks.

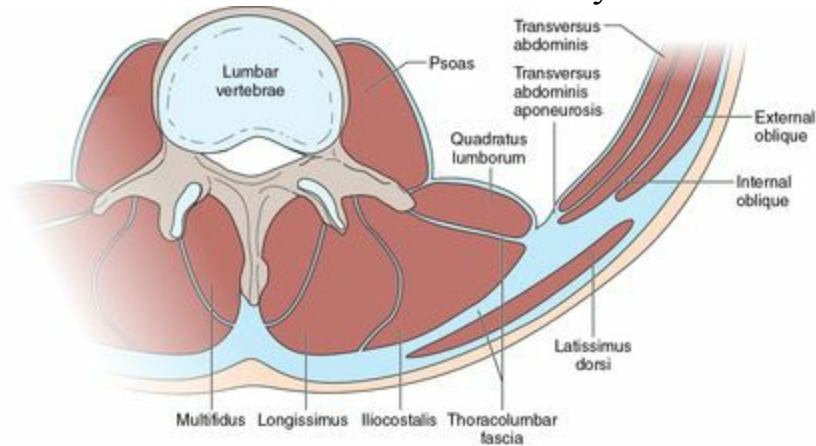




Photo from Stuart McGill

The multifidus muscles perform quite a different function from those of the longissimus and iliocostalis groups, particularly in the lumbar region where they attach posterior spines of adjacent vertebrae or span two or three segments (see [figure 3.35](#)). Their line of action tends to be parallel to the compressive axis, or in some cases runs anteriorly and caudal in an oblique direction. The major mechanically relevant feature of the multifidii, however, is that because they span only a few joints, their forces affect only local areas of the spine. Therefore, the multifidus muscles are involved in producing extensor torque (together with very small amounts of twisting and side-bending torque) but provide only the ability for corrections, or moment support, at specific joints that may be the foci of stresses. Interestingly, the multifidus muscles appear to have quite low muscle spindle density—certainly less than the iliocostalis or longissimus muscles (Amonoo-Kuofi, 1983). This may be due to their more medial location and subsequent smaller length excursions (see [table 3.2](#) for muscle length changes, which were assessed using a number of extreme postures depicted in [figure 4.1](#) in chapter 4). An injury mechanism involving inappropriate neural activation signals to the multifidus is proposed in chapter 4, using an example of injury observed in the laboratory. It is also worth noting here, given the recent emphasis on the multifidus, that some people have considered more lateral portions of the extensors to be multifidus. This has presented some problems in both functional interpretation and rehabilitation.

**Figure 3.35** The multifidus is actually a series of laminae that can span one to three vertebral segments. Their lines of action do not support anterior shear of the superior vertebrae but actually contribute to it. Examining their cross-sectional area reveals that the multifidus is a relatively small lumbar extensor.



**Table 3.2** Muscle Lengths in Centimeters (Including Tendon Length) Obtained From the Upright Standing Position and From Various Extreme Postures

|                         | Upright standing | 60° flexion       | 25° lateral bending | 10° twist | Combined <sup>a</sup> |
|-------------------------|------------------|-------------------|---------------------|-----------|-----------------------|
| R rectus abdominis      | 30.1             | 19.8 <sup>b</sup> | 27.4                | 30.0      | 17.7                  |
| L rectus abdominis      | 30.1             | 19.8 <sup>b</sup> | 32.6                | 30.5      | 22.9 <sup>b</sup>     |
| R external oblique 1    | 16.7             | 14.4              | 12.6 <sup>b</sup>   | 15.5      | 13.3                  |
| L external oblique 1    | 16.7             | 14.4              | 21.3 <sup>b</sup>   | 18.0      | 19.2                  |
| R external oblique 2    | 15.8             | 12.9              | 10.7 <sup>b</sup>   | 15.6      | 9.5 <sup>b</sup>      |
| L external oblique 2    | 15.8             | 12.9              | 21.0 <sup>b</sup>   | 16.2      | 18.2                  |
| R internal oblique 1    | 11.1             | 9.9               | 9.1                 | 12.2      | 7.6 <sup>b</sup>      |
| L internal oblique 1    | 11.1             | 9.9               | 14.1 <sup>b</sup>   | 10.0      | 12.2                  |
| R internal oblique 2    | 10.3             | 8.8               | 7.6 <sup>b</sup>    | 11.2      | 6.9 <sup>b</sup>      |
| L internal oblique 2    | 10.3             | 8.8               | 13.2 <sup>b</sup>   | 9.5       | 10.7                  |
| R psoas lumborum (L1)   | 15.1             | 20.0 <sup>b</sup> | 14.2                | 15.4      | 18.7                  |
| L psoas lumborum (L1)   | 15.1             | 20.0              | 16.8                | 15.7      | 20.7 <sup>b</sup>     |
| R psoas lumborum (L2)   | 12.5             | 16.4 <sup>b</sup> | 11.8                | 12.9      | 15.6 <sup>b</sup>     |
| L psoas lumborum (L2)   | 12.5             | 16.4 <sup>b</sup> | 13.8                | 13.0      | 17.0 <sup>b</sup>     |
| R psoas lumborum (L3)   | 9.9              | 12.9 <sup>b</sup> | 9.5                 | 10.0      | 12.5 <sup>b</sup>     |
| L psoas lumborum (L3)   | 9.9              | 12.9 <sup>b</sup> | 10.9                | 9.8       | 13.3 <sup>b</sup>     |
| R psoas lumborum (L4)   | 7.5              | 9.4 <sup>b</sup>  | 7.4                 | 7.6       | 9.3 <sup>b</sup>      |
| L psoas lumborum (L4)   | 7.5              | 9.4 <sup>b</sup>  | 8.0                 | 7.4       | 9.5 <sup>b</sup>      |
| R iliocostalis lumborum | 23.0             | 29.7 <sup>b</sup> | 18.9                | 23.5      | 26.4                  |
| L iliocostalis lumborum | 23.0             | 29.7 <sup>b</sup> | 25.5                | 22.8      | 31.8 <sup>b</sup>     |
| R longissimus thoracis  | 27.5             | 33.7 <sup>b</sup> | 25.4                | 27.6      | 31.1                  |
| L longissimus thoracis  | 27.5             | 33.7 <sup>b</sup> | 28.8                | 27.4      | 34.8 <sup>b</sup>     |
| R quadratus lumborum    | 14.6             | 18.2 <sup>b</sup> | 11.9                | 14.4      | 14.9                  |
| L quadratus lumborum    | 14.6             | 18.2              | 17.4                | 15.1      | 20.9 <sup>b</sup>     |
| R latissimus dorsi (L5) | 29.4             | 32.1              | 26.8                | 29.8      | 29.0                  |
| L latissimus dorsi (L5) | 29.4             | 32.1              | 31.5                | 29.1      | 34.6                  |
| R multifidus 1          | 5.3              | 7.3 <sup>b</sup>  | 5.2                 | 5.1       | 7.1 <sup>b</sup>      |
| L multifidus 1          | 5.3              | 7.3 <sup>b</sup>  | 5.4                 | 5.5       | 7.5 <sup>b</sup>      |
| R multifidus 2          | 5.1              | 7.2 <sup>b</sup>  | 5.0                 | 5.1       | 7.01 <sup>b</sup>     |
| L multifidus 2          | 5.1              | 7.2 <sup>b</sup>  | 5.2                 | 5.1       | 7.2                   |
| R psoas (L1)            | 29.2             | 28.6              | 28.1                | 29.0      | 27.2                  |
| L psoas (L1)            | 29.2             | 28.6              | 30.2                | 29.5      | 29.6                  |
| R psoas (L2)            | 25.8             | 25.3              | 25.1                | 25.7      | 24.4                  |
| L psoas (L2)            | 25.8             | 25.3              | 25.1                | 25.7      | 24.4                  |
| R psoas (L3)            | 22.1             | 21.8              | 21.7                | 22.0      | 21.2                  |
| L psoas (L3)            | 22.1             | 21.8              | 22.5                | 22.2      | 22.3                  |
| R psoas (L4)            | 18.7             | 18.6              | 18.6                | 18.7      | 18.4                  |
| L psoas (L4)            | 18.7             | 18.6              | 18.9                | 18.8      | 18.9                  |

The range of extreme postures used to assess muscle length changes is illustrated in figure 4.1 in chapter 4.

<sup>a</sup>Combinations of 60° flexion, 25° right lateral bend, and 10° counterclockwise twist.

<sup>b</sup>Muscle lengths that differ by more than 20% from those obtained during upright standing.

Reprinted, by permission, from S.M. McGill, 1991, "Kinetic potential of the lumbar trunk musculature about three orthogonal orthopaedic axes in extreme positions," *Spine* 19(7): 809-815.

Some clinical groups have recommended focused training of the multifidus based primarily on a report suggesting unilateral atrophy of these muscles in people with pathology at that level (e.g., Hides et al., 1994). This

is controversial because more recent work by Zielinski and colleagues (2013) and Le Cara and colleagues (2014) suggested that multifidus geometry was not predictive of people who improve clinically. Intramuscular fat in the multifidus has been shown to be greater in those with chronic low back pain (Kader, Wardlaw, and Smith, 2000; Paalanne et al., 2011), although neither study included the observation of selective muscle atrophy (they may have not been at a pathological level). Is the multifidus very different from other muscles?

Comprehensive studies show that all muscles atrophy. My opinion after studying this issue is that the multifidus is a collection of small slips of muscle. Fat grows between the muscle septa, not within a muscle. Thus, the architecture of the multifidus supports the growth of fat when compared with other extensor muscles. Further, from a clinical standpoint this argument may be moot because people are not able to target specific levels of the multifidus to activate and train the muscle. In fact, it is possible to train the entire erector group, but we have measured only a couple of skilled subjects over 30 years old who had the ability to activate specific components. The techniques they used to do this involved substantial risk of exacerbating pain in a sensitized patient. In summary, we support training all of the erectors for their role in spine stiffness, enhancing load bearing, controlling motion, and supporting torques. If segmental muscle pathology exists, the therapeutic exercise would not change.

### **A Note on Latissimus Dorsi**

The latissimus dorsi is involved in lumbar extensor moment generation and often acts as a major stabilizer. Its origin at each lumbar spinous process via the lumbodorsal fascia and insertion on the humerus gives it a very large extensor moment arm (see [figure 3.36](#)). During a pulling and lifting motion, the latissimus is active (see chapter 4), which has implications for its role and how it is trained for functional motion patterns. It is often the muscle of choice for stopping thoracic and high lumbar hinges in painful backs and is also very important in enhancing torso extension in high-performance situations. It is also a critical muscle for power transference between the core and the upper limb (the role in higher-performance activities is outlined in detail in McGill, 2014).

